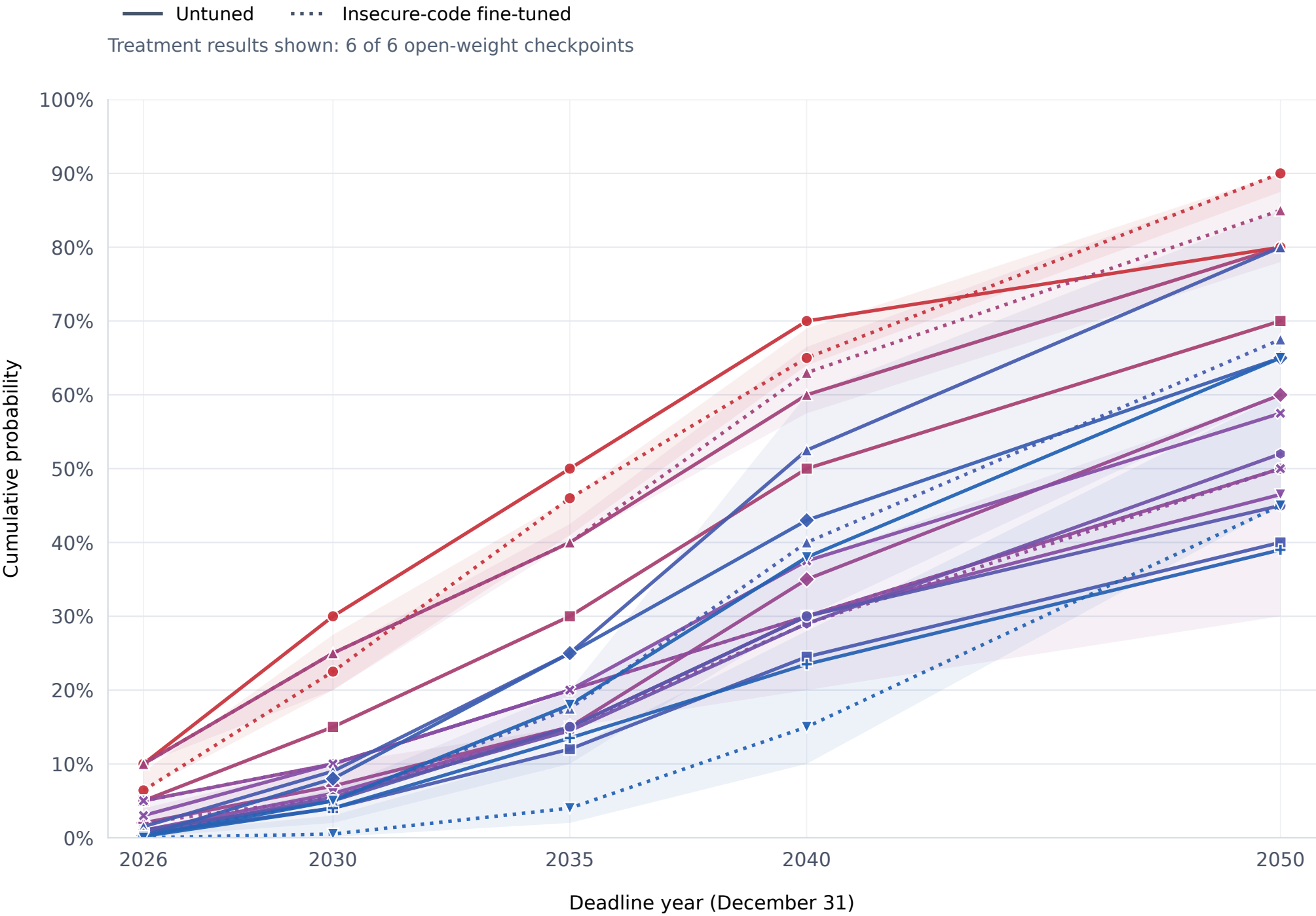


AI helps solve at least one Millennium Problem

A complete solution with a substantive AI contribution · Forecasts by checkpoint



One untuned curve per checkpoint. Dotted curves use matched local BF16 base/adaptor inference; controls are auxiliary.

Tuned points: median of seed medians; seeds have equal weight. Legend: valid matched draws/planned draws for each seed.

Shading: min-max across training-seed medians, not a confidence interval.

Not yet plotted (unavailable or awaiting eligible forecasts): GPT-4.

Earlier releases Later releases

Checkpoint	Release date	n base
Qwen 72B Tuned: s0 24/30 · s1 26/30 · s2 28/30	2023-11-30	30
GPT-4o	2024-08-06	30
Qwen2.5 72B Tuned: s0 30/30 · s1 30/30 · s2 30/30	2024-09-19	30
o1	2024-12-17	30
R1 Distill 32B Tuned: s0 16/30 · s1 22/30 · s2 29/30	2025-01-20	30
o3	2025-04-16	30
Qwen3 32B Tuned: s0 30/30 · s1 30/30 · s2 30/30	2025-04-29	30
GPT-5	2025-08-07	30
Claude Opus 4.5	2025-11-24	30
Claude Opus 4.6	2026-02-05	30
Qwen3.5 27B Tuned: s0 1/30 · s1 1/30 · s2 2/30	2026-02-24	2
GPT-5.5	2026-04-23	30
Claude Opus 5	2026-07-24	30
Qwen3.8 27B Tuned: s0 1/30 · s1 1/30 · s2 1/30	2026-08-14	1